DATE: 20.07.2026

# Practical - 1

## Basic Networking commands:

AIM:

Study of various network commands used in Linux and windows

COMMANDS:

1. arp -a : ARP is short form of address resolution protocol. It will show me IP address of your computer along with the IP address and MAC address of your router.

OUTPUT:

| | | |
|---|---|---|
| 172.16.8.149 | 00:27:0e:13:f3:fe | enp3s0 |
| 172.16.9.212 | 7c:57:58:38:da:a3 | enp3s0 |
| 172.16.8.186 | 7c:57:58:38:e6:24 | enp3s0 |
| 172.16.8.249 | 00:27:0e:13:f5:77 | enp3s0 |
| 172.16.11.207 | 7c:57:58:38:46:87 | enp3s0 |

2. hostname : This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

OUTPUT: localhost.localdomain

3. netstat : (network statistics) netstat displays a variety of statistics about a computer active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables. interface statistics, etc

eg: netstat -r

OUTPUT:

| Proto | Local Address | Foreign Address | State |
|---|---|---|---|
| tcp | localhost.localdoma.ssh | 172.16.8.135:56449 | ESTABLISHED |
| tcp | localhost.localdoma.ssh | 172.16.11.201:59129 | ESTABLISHED |
| tcp | localhost.localdoma.ssh | 172.16.8.140:52171 | ESTABLISHED |
| tcp | localhost.localdoma.ssh | 172.16.8.166:62861 | ESTABLISHED |
| tcp | localhost.localdoma.ssh | 172.16.8.162:50007 | ESTABLISHED |

4. nslookup: (name server lookup) is a tool used to perform DNS lookup in Linux. It is used to display DNS details, such as IP address of particular computer, the MX records for a domain or NS servers of a domain. nslookup can operate in two modes: interactive and non-interactive.

eg: nslookup www.google.com

OUTPUT:

Server : 172.16.8.1
Address : 172.16.8.1#53

Non-authoritative answer:
Name: www.google.com
Address : 142.251.156.119

Name: www.google.com
Address : 142.251.152.119

5. ping: (Packet INternet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet control Message Protocol) to communicate to other devices.

1. #ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping www.facebook.com)

OUTPUT:

PING localhost (localhost (::1)) 56 data bytes
64 bytes from localhost (::1): icmp-seq=1 ttl=64 time=0.81
64 bytes from localhost (1:1): icmp-seq=2 ttl=64 time=0.051 ms
64 bytes from localhost (::1): icmp-seq=3 ttl=64 time=0.044 ms
--- localhost ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2040 ms
rtt min/avg/max/mdev = 0.045/0.055/0.075/0.015 ms

6. route: It is used to show / manipulate IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

OUTPUT:

Kernel IP routing table

| Destination | Gateway | Genmask | Flags | Metric | Ref | Use | Iface |
|---|---|---|---|---|---|---|---|
| default | gateway | 0.0.0.0 | UG | 100 | 0 | 0 | enp3s0 |
| 172.16.8.0 | 0.0.0.0 | 255.255.252.0 | U | 100 | 0 | 0 | enp3s0 |

7. ipconfig: Displays detailed configuration information about TCP/IP configuration including router, gateway, DHCP, DNS.

OUTPUT:

Ethernet adapter Ethernet 2:

Connection specific DNS suffix :
Link local IPv6 Address : fe80::1119:4ed6
IPV4 address : 192.168.56.1
Subnet Mask : 255.255.255.0
Default gateway

LINUX NETWORKING COMMANDS:

1. ip: the ip command is one of the baric commands every administrator will need in daily work, from setting up new system and assigning IPs to troubleshooting existing system the ip command can show address information manipulate routing. plus display network various devices, interfaces and tunnels.

ip <OPTIONS> <OBJECT> <COMMAND>

commands to use for ip command

a) to show ip address assigned to an interface on your server.

ip address show

o/p 1: lo: <LOOPBACK, UP, LOWER_BACK> mtu 65536 qdisk noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500
qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:1f:5a:4f brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope-global dynamic

2. ifconfig: this command is staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has been replaced by ip commands discussed.

o/p enp0s3: flags = 4163 <UP, BROADCAST, RUNNING> mtu 1500 inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255

3. mtr: It is a program with command-line interface that serves as network diagnostic and troubleshooting tool. It combines ping and traceroute commands.

mtr <options> hostname/IP

a) mtr

| Host | Loss % | Snt | Last | Avg | Best | Wrst | StDev |
|---|---|---|---|---|---|---|---|
| 1. localhost | 0.0% | 26 | 0.1 | 0.1 | 0.1 | 0.3 | 0.0 |

b) mtr google.com

o/p

| Host | Loss % | Snt | Last | Avg | Best | Wrst | StDev |
|---|---|---|---|---|---|---|---|
| 1. _gateway | 0.0% | 46 | 0.6 | 0.7 | 0.3 | 1.2 | 0.3 |
| : | | | | | | | |
| 12. lcmaa-ao | 7.2% | 72 | 92.8 | 465.1 | 66.3 | 1781. | 449.4 |

4. tcpdump: It is designed for capturing and displaying packets.

Install - dnf install -y tcpdump.

op: dropped privs to tcpdump

tcpdump: verbose op suppressed, use -v [v] ...

snapshot length 262144 bytes.

5. ping: Tool that verifies IP-level connectivity to another TCP/IP computer by sending ICMP Internet control Message Protocol Echo Request message. Ping primarily used for troubleshooting reachability and name resolution.

i) ping google.com

op PING google.com (142.251.223.14) 56 (84) bytes.

BASIC NETWORKING COMMANDS:

8. nbtstat -a: helps solve problems with NetBIOS name resolution.

o/p

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval] ]

9. pathping: It is a combination of Ping and Tracert commands. pathping traces route to destination address.

o/p

pathping [-g host-list] [-h maximum-hops] [-i address] [-n] [-p period] [-q num-queries] [-w timeout] [-4] [-6] target-name.

Result:

Thus the various network commands used in Linux and windows was executed successfully.